EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 05

Trigonometric Functions

1 Concepts | 82 questions

B5 Landscape | one question per teaching page

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Chapter 05 | Trigonometric Functions

CLASSROOM ROUTE

Concept 1 | Defining Angles and Basic Trigonometric Functions

82 questions

Official lessons: 5-1, 5-2, 5-3, 5-4

The Formula Bank changes at each Concept boundary.

Concept 1 | Defining Angles and Basic Trigonometric Functions — **OFFICIAL LESSONS 5-1, 5-2, 5-3, 5-4**

Defining Angles and Basic Trigonometric Functions

Equation-first recall | use the same rail beside every question in this Concept

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Start with the family cue, write the first move, then use the working grid.

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V01

Q001 | WRITTEN

C05-K01-5-1-WU1

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus 75°; (b) two full clockwise revolutions and a further 135°.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q002 | WRITTEN

C05-K01-5-1-WU2

Draw the terminal side for 300° and −450°.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q003 | WRITTEN

C05-K01-5-1-WU3

Express 420° and −210° as $\alpha + 360° \times n$, where n is an integer and $0° \leq \alpha < 360°$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

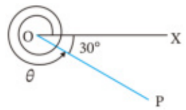
Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V01

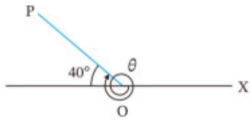
Q004 | WRITTEN C05-K01-5-1-TR1

Find the measures of θ in the two figures: (a) a clockwise turn of 30° from the positive x-axis; (b) a terminal side 40° above the negative x-axis, measured counter-clockwise from the positive x-axis).

(a)



(b)



WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q005 | WRITTEN

C05-K01-5-1-TR2

Draw the terminal side for $500°$ and $-120°$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q006 | WRITTEN C05-K01-5-1-TR3

Express 520° and −140° as $\alpha + 360° \times n$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

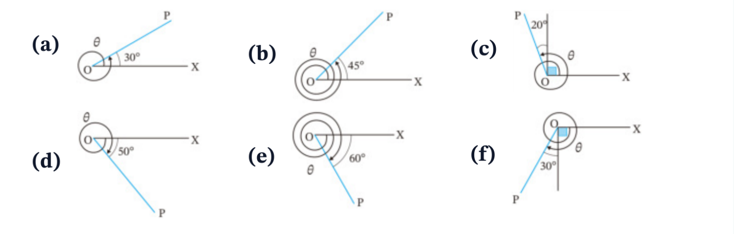
Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V01

Q007 | WRITTEN

C05-K01-5-1-EX1

Find the six diagram measures θ shown in the source figure (a)–(f).



WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q008 | WRITTEN

C05-K01-5-1-EX2

Draw the terminal sides for $470°,\ 780°,\ 225°,\ -380°,\ -990°,\ -200°$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q009 | WRITTEN

C05-K01-5-1-EX3

Express $610°,\ 920°,\ 840°,\ -430°,\ -960°,\ -20°$ as $\alpha + 360° \times n$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q010 | WRITTEN

C05-K01-5-1-CH1

Express $129720°$ and $-2592100°$ as $\alpha + 360° \times n$, with $0° \leq \alpha < 360°$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q011 | MCQ

C05-K01-5-1-GM01

The principal angle coterminal with 765° is:

A 45°	**B** 135°
C 225°	**D** 315°

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q012 | MCQ

C05-K01-5-1-GM02

The principal angle coterminal with $-410°$ is:

A $-50°$	**B** $50°$
C $310°$	**D** $410°$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q013 | MCQ

C05-K01-5-1-GM03

The terminal side of 540° lies on the:

A positive x-axis

B negative x-axis

C positive y-axis

D negative y-axis

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q014 | MCQ

C05-K01-5-1-GM04

A positive angle of 315° terminates in:

A QI

B QII

C QIII

D QIV

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q015 | MCQ C05-K01-5-1-GM05

Which is the general-angle form for $-75°$?:

A $75° + 360°n$

B $285° + 360°n$

C $-285° + 360°n$

D $360°n - 75°$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V01

Q016 | MCQ

C05-K01-5-1-GM06

The direction of a negative angle is:

A counter-clockwise

B clockwise

C radial only

D undefined

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q017 | MCQ

C05-K01-5-1-GM07

The terminal side of $1080°$ is the:

A positive x-axis

B negative x-axis

C positive y-axis

D negative y-axis

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q018 | MCQ C05-K01-5-1-GM08

For $\alpha = 0°$, the general-angle family is:

A $360°n$

B $0°n$

C $180° + 360°n$

D $90° + 360°n$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q019 | MCQ

C05-K01-5-1-GM09

The principal angle for -990° is:

A 90°

B 180°

C 270°

D -270°

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V01

Q020 | MCQ

C05-K01-5-1-GM10

A ray (P) is 25° above the negative x-axis. Its standard-position angle is:

A 25°

B 155°

C 205°

D 335°

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q021 | WRITTEN

C05-K01-5-2-WU

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q022 | WRITTEN

C05-K01-5-2-TR

Complete the standard-angle degree/radian table, then: convert $288°,\ -400°$ to radians; convert $\frac{11\pi}{6},\ -\frac{3\pi}{5}$ to degrees; and find $L,\ S$ for $r = 6,\ \theta = \frac{3\pi}{4}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q023 | WRITTEN

C05-K01-5-2-EX1

Convert $108°,\ 36°,\ -240°,\ -225°,\ 420°,\ 400°,\ -570°,\ -2700°$ to radians.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V02

Q024 | WRITTEN

C05-K01-5-2-EX2

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ to degrees.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Q025 | WRITTEN

C05-K01-5-2-EX3

Find L and S for: (a) $r = 6,\ \theta = \frac{7\pi}{6}$; (b) $r = 4,\ \theta = \frac{2\pi}{5}$; (c) $r = 6,\ \theta = \frac{5\pi}{4}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V04

Q026 | WRITTEN

C05-K01-5-2-EX4

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V04

Q027 | WRITTEN

C05-K01-5-2-CH1

A sector of radius 9 cm and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q028 | MCQ

C05-K01-5-2-GM01

Convert 150° to radians:

A $\frac{5\pi}{6}$

B $\frac{3\pi}{5}$

C $\frac{5\pi}{3}$

D $\frac{\pi}{6}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V02

Q029 | MCQ

C05-K01-5-2-GM02

Convert $-\frac{5\pi}{6}$ to degrees:

A $-120°$

B $-150°$

C $150°$

D $-300°$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Q030 | MCQ

C05-K01-5-2-GM03

For $r = 5,\ \theta = \frac{2\pi}{3}$, the arc length is:

A $\frac{5\pi}{3}$

B $\frac{10\pi}{3}$

C $\frac{25\pi}{3}$

D 10π

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Q031 | MCQ

C05-K01-5-2-GM04

For $r = 4,\ \theta = \frac{\pi}{2}$, the sector area is:

A π

B 2π

C 4π

D 8π

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q032 | MCQ

C05-K01-5-2-GM05

The radian measure of a full turn is:

A π

B 2π

C 180

D 360

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V02

Q033 | MCQ

C05-K01-5-2-GM06

The degree measure of $\frac{7\pi}{4}$ is:

A 210°

B 270°

C 315°

D 360°

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V04

Q034 | MCQ

C05-K01-5-2-GM07

The principal angle for $-\frac{11\pi}{6}$ is:

A $\frac{\pi}{6}$

B $-\frac{\pi}{6}$

C $\frac{5\pi}{6}$

D $\frac{11\pi}{6}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Q035 | MCQ

C05-K01-5-2-GM08

For $r = 3,\ \theta = \pi$, the sector area is:

A $\frac{3\pi}{2}$

B 3π

C 9π

D $\frac{9\pi}{2}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V04

Q036 | MCQ C05-K01-5-2-GM09

The angle $\frac{5\pi}{3}$ lies in:

A QI

B QII

C QIII

D QIV

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q037 | WRITTEN

C05-K01-5-2-GW01

Convert 252° to radians.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Q038 | WRITTEN C05-K01-5-2-GW02

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q039 | WRITTEN

C05-K01-5-3-WU

If $\theta = \frac{4\pi}{3}$, find $\sin\theta,\ \cos\theta,\ \tan\theta$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q040 | WRITTEN C05-K01-5-3-TR

Find the exact values (write undefined where appropriate): $\sin\frac{7\pi}{4}$, $\cos\frac{\pi}{3}$, $\tan\frac{\pi}{6}$, $\sin(-\frac{2\pi}{3})$, $\cos(-\pi)$, $\tan(-\frac{\pi}{2})$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q041 | WRITTEN

C05-K01-5-3-EX

Find the exact values of the eighteen trigonometric functions listed in the source exercise (a)–(r), including negative and undefined cases.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V02

Q042 | WRITTEN

C05-K01-5-3-CH1

The panel height is $h(t) = 3\sin(\frac{2\pi}{3}t - \frac{\pi}{4}) + 1.5$. Find the exact height at $t = 0$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q043 | MCQ

C05-K01-5-3-GM01

$\sin \frac{\pi}{6}$ equals:

A 0

B $\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D 1

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q044 | MCQ

C05-K01-5-3-GM02

$\cos\frac{2\pi}{3}$ equals:

A -1

B $-\frac{1}{2}$

C $\frac{1}{2}$

D $\frac{\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q045 | MCQ

C05-K01-5-3-GM03

$\tan\frac{3\pi}{4}$ equals:

A -1

B 0

C 1

D undefined

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V02

Q046 | MCQ

C05-K01-5-3-GM04

$\sin(-\frac{\pi}{2})$ equals:

A -1

B 0

C 1

D undefined

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q047 | MCQ

C05-K01-5-3-GM05

$\tan\frac{\pi}{2}$ is:

A 0	**B** 1
C -1	**D** undefined

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q048 | MCQ

C05-K01-5-3-GM06

$\cos \frac{5\pi}{3}$ equals:

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q049 | MCQ

C05-K01-5-3-GM07

If $P(x,\ y)$ lies on the unit circle, then $\sin\theta$ is:

A x

B y

C x/y

D 1/x

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V02

Q050 | MCQ

C05-K01-5-3-GM08

$\tan(-\frac{3\pi}{4})$ equals:

A -1

B 0

C 1

D undefined

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q051 | MCQ

C05-K01-5-3-GM09

$\sin \frac{11\pi}{6}$ equals:

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q052 | WRITTEN

C05-K01-5-3-GW01

Find $\sin\frac{5\pi}{6}$, $\cos\frac{4\pi}{3}$, $\tan\frac{7\pi}{4}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V02

Q053 | WRITTEN C05-K01-5-3-GW02

Find $\sin(-\frac{5\pi}{6}),\ \cos(-\frac{7\pi}{6}),\ \tan(-\frac{\pi}{3})$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q054 | WRITTEN C05-K01-5-3-GW03

For $P = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ on the unit circle, find the corresponding $\sin\theta,\ \cos\theta,\ \tan\theta$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q055 | WRITTEN C05-K01-5-3-GW04

Find the exact value of $3\sin\frac{3\pi}{2} - 2\cos\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q056 | WRITTEN

C05-K01-5-3-GW05

State the two standard angles where $\tan\theta$ is undefined in $[0,\ 2\pi)$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q057 | WRITTEN

C05-K01-5-4-WU

Solve for θ, $0 \le \theta < 2\pi$: (a) $\sin\theta = -\frac{1}{2}$; (b) $\cos\theta = \frac{1}{\sqrt{2}}$; (c) $\tan\theta = -\sqrt{3}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q058 | WRITTEN C05-K01-5-4-TR

Solve the six trigonometric equations in the source Try set for $0 \le \theta < 2\pi$: $\sin\theta = 1/\sqrt{2}$, $\cos\theta = -\sqrt{3}/2$, $\tan\theta = 1/\sqrt{3}$, $\sin\theta = 0$, $\cos\theta = 1/2$, $\tan\theta = -1$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q059 | WRITTEN

C05-K01-5-4-EX

Solve the sixteen equations in the source Exercise set (a)–(p) for $0 \leq \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V03

Q060 | WRITTEN C05-K01-5-4-CH1

Solve $\tan^2\theta + (1-\sqrt{3})\tan\theta = \sqrt{3}$ for $0 \le \theta < 360°$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q061 | MCQ   C05-K01-5-4-GM01

For $0 \le \theta < 2\pi$, $\sin\theta = \frac{1}{2}$ gives:

A $\frac{\pi}{6}, \frac{5\pi}{6}$

B $\frac{\pi}{3}, \frac{2\pi}{3}$

C $\frac{\pi}{6}, \frac{7\pi}{6}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q062 | MCQ C05-K01-5-4-GM02

For $0 \leq \theta < 2\pi$, $\cos\theta = -\frac{1}{2}$ gives:

A $\frac{\pi}{3}, \frac{5\pi}{3}$

B $\frac{2\pi}{3}, \frac{4\pi}{3}$

C $\frac{\pi}{6}, \frac{5\pi}{6}$

D $\frac{4\pi}{3}, \frac{5\pi}{3}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q063 | MCQ

C05-K01-5-4-GM03

For $0 \leq \theta < 2\pi$, $\tan\theta = 1$ gives:

A $\frac{\pi}{4}, \frac{5\pi}{4}$

B $\frac{3\pi}{4}, \frac{7\pi}{4}$

C $\frac{\pi}{4}, \frac{7\pi}{4}$

D $\frac{3\pi}{4}, \frac{5\pi}{4}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q064 | MCQ

C05-K01-5-4-GM04

The solutions of $\sin\theta = 0$ in $[0,\ 2\pi)$ are:

A $0,\ \pi$

B $\frac{\pi}{2}, \frac{3\pi}{2}$

C $0,\ 2\pi$

D π

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q065 | MCQ

C05-K01-5-4-GM05

If $\cos\theta = 1$, then in $[0,\ 2\pi)$, θ is:

A 0

B $\frac{\pi}{2}$

C π

D $\frac{3\pi}{2}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q066 | MCQ C05-K01-5-4-GM06

For $0 \le \theta < 2\pi$, $\tan\theta = -\sqrt{3}$ gives:

A $\frac{\pi}{3}, \frac{4\pi}{3}$

B $\frac{2\pi}{3}, \frac{5\pi}{3}$

C $\frac{\pi}{6}, \frac{7\pi}{6}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q067 | MCQ

C05-K01-5-4-GM07

For $0 \le \theta < 2\pi$, $\sin\theta = -1$ gives:

A $\frac{\pi}{2}$

B $\frac{3\pi}{2}$

C π

D 0

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q068 | MCQ C05-K01-5-4-GM08

For $0 \le \theta < 2\pi$, $\cos\theta = 0$ gives:

A $0, \pi$

B $\frac{\pi}{2}, \frac{3\pi}{2}$

C π

D $\frac{\pi}{4}, \frac{5\pi}{4}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q069 | MCQ

C05-K01-5-4-GM09

The equation $\tan\theta = 0$ has solutions in $[0,\ 2\pi)$:

A $0,\ \pi$

B $\frac{\pi}{2},\ \frac{3\pi}{2}$

C $\frac{\pi}{4},\ \frac{5\pi}{4}$

D $\pi,\ 2\pi$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q070 | WRITTEN

C05-K01-5-4-GW01

Solve $\cos\theta = \frac{\sqrt{3}}{2}$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q071 | WRITTEN

C05-K01-5-4-GW02

Solve $\sin\theta = -\frac{\sqrt{3}}{2}$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q072 | WRITTEN

C05-K01-5-4-GW03

Solve $\tan\theta = \frac{1}{\sqrt{3}}$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q073 | WRITTEN

C05-K01-5-4-GW04

Solve $2\cos\theta - 1 = 0$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q074 | WRITTEN

C05-K01-5-4-GW05

Solve $2\sin\theta + \sqrt{2} = 0$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Quiz M1 | QUIZ MCQ C05-K01-Q-M01

The radian measure of 240° is:

A $\frac{2\pi}{3}$

B $\frac{4\pi}{3}$

C $\frac{3\pi}{4}$

D $\frac{5\pi}{3}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Quiz M2 | QUIZ MCQ

C05-K01-Q-M02

If $\sin\theta = -\frac{\sqrt{3}}{2}$, $0 \le \theta < 2\pi$, then θ is:

A $\frac{\pi}{3}, \frac{2\pi}{3}$

B $\frac{4\pi}{3}, \frac{5\pi}{3}$

C $\frac{2\pi}{3}, \frac{4\pi}{3}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Quiz M3 | QUIZ MCQ | C05-K01-Q-M03

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

A 3π	**B** 6π
C 12π	**D** 18π

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Quiz M4 | QUIZ MCQ

C05-K01-Q-M04

The general-angle family for $-300°$ is:

A $60° + 360° n$

B $300° + 360° n$

C $-60° + 360° n$

D $120° + 360° n$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$\sin\theta = y$ $\quad \cos\theta = x$ $\quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Quiz W1 | QUIZ WRITTEN

C05-K01-Q-W01

Solve $\cos\theta = -\frac{\sqrt{2}}{2}$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V02

Quiz W2 | QUIZ WRITTEN

C05-K01-Q-W02

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Quiz W3 | QUIZ WRITTEN

C05-K01-Q-W03

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Quiz W4 | QUIZ WRITTEN

C05-K01-Q-W04

Evaluate $3\sin\frac{7\pi}{6} - 2\cos\frac{4\pi}{3}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 05

Trigonometric Functions

1 Concepts | 62 synchronized questions

B5 Landscape | one question per teaching page

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Chapter 05 | Trigonometric Functions

CLASSROOM ROUTE

Concept 2 | Trigonometric Identities and Expression Evaluation

62 questions

Official lessons: 5-5, 5-6, 5-7

The Formula Bank changes automatically at each Concept boundary. Every question ID and order remains synchronized with the approved A4 Classified and Solutions pair.

Concept 2 | Trigonometric Identities and Expression Evaluation OFFICIAL LESSONS 5-5, 5-6, 5-7

Trigonometric Identities and Expression Evaluation

Equation-first recall | use the same rail beside every question in this Concept

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Start with the family cue, write the first move, then use the working grid.

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q001 | WRITTEN C05-K02-5-5-WU1

Given (a) $\sin\theta = -\frac{1}{5}$ in QIII and (b) $\tan\theta = -\sqrt{6}$ with $\frac{3\pi}{2} < \theta < 2\pi$, find the other two trigonometric values in each case.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q002 | WRITTEN C05-K02-5-5-TR1

Given one value and the quadrant, find the other two: (a) $\cos\theta = -1/\sqrt{3}$, QIII; (b) $\tan\theta = -2$, QIV; (c) $\sin\theta = 3/5$, QII; (d) $\tan\theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V02

Q003 | WRITTEN

C05-K02-5-5-EX1

For each of the twelve numbered cases in the source Exercise, one of $\sin\theta, \cos\theta, \tan\theta$ and a quadrant/interval is given. Find the other two values.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V03

Q004 | WRITTEN

C05-K02-5-5-CH1

Using trigonometric identities, find the other two values given $\tan\theta + \cot\theta = 12$ and $\pi < \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q005 | WRITTEN

C05-K02-5-5-GW01

Given $\sin\theta = \dfrac{5}{13}$ in QII, find $\cos\theta$ and $\tan\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q006 | WRITTEN   C05-K02-5-5-GW02

Given $\cos\theta = -\frac{3}{5}$ in QIII, find $\sin\theta$ and $\tan\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q007 | WRITTEN C05-K02-5-5-GW03

Given $\tan\theta = \frac{3}{4}$ in QI, find $\sin\theta$ and $\cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q008 | WRITTEN

C05-K02-5-5-GW04

Given $\tan\theta = -\dfrac{5}{12}$ in QIV, find $\sin\theta$ and $\cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q009 | WRITTEN

C05-K02-5-5-GW05

Given $\sin\theta = -\frac{\sqrt{2}}{2}$ in QIII, find $\cos\theta$ and $\tan\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q010 | MCQ

C05-K02-5-5-GM01

If $\sin\theta = -\frac{3}{5}$ in QIII, $\cos\theta$ is:

A $\frac{4}{5}$

B $-\frac{4}{5}$

C $\frac{3}{5}$

D $-\frac{3}{5}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q011 | MCQ

C05-K02-5-5-GM02

If $\tan\theta = 2$ in QI, $\sin\theta$ is:

A $\frac{2\sqrt{5}}{5}$

B $\frac{\sqrt{5}}{5}$

C $-\frac{2\sqrt{5}}{5}$

D $\frac{2}{5}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q012 | MCQ

C05-K02-5-5-GM03

If $\cos\theta = \frac{1}{\sqrt{2}}$ in QIV, $\tan\theta$ is:

A 1

B -1

C $\sqrt{2}$

D $-\sqrt{2}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V02

Q013 | MCQ

C05-K02-5-5-GM04

For $\tan\theta = -\sqrt{3}$ in QII, $\sin\theta$ is:

A $\frac{1}{2}$

B $-\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D $-\frac{\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q014 | MCQ

C05-K02-5-5-GM05

If $\sin\theta = \frac{7}{25}$ in QI, $\tan\theta$ is:

A $\frac{7}{24}$

B $\frac{24}{7}$

C $-\frac{7}{24}$

D $\frac{25}{7}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q015 | WRITTEN C05-K02-5-6-WU

Given $\sin\theta + \cos\theta = \frac{1}{2}$, with θ in QII, evaluate (a) $\sin\theta\cos\theta$, (b) $\sin^3\theta + \cos^3\theta$, (c) $\sin\theta - \cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q016 | WRITTEN　　C05-K02-5-6-TR

Given $\sin\theta + \cos\theta = \frac{1}{3}$, with θ in QII, evaluate (a) $\sin\theta\cos\theta$, (b) $\sin^3\theta + \cos^3\theta$, (c) $\sin\theta - \cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q017 | WRITTEN

C05-K02-5-6-EX1

Solve the three numbered Exercise problems: (1) given $\sin\theta - \cos\theta = \sqrt{3}/2$ in QI; (2) given $\sin\theta + \cos\theta = \sqrt{3}/2$ in QIV; (3) given $\sin\theta - \cos\theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V03

Q018 | WRITTEN

C05-K02-5-6-CH1

Given $\sin\theta - \cos\theta = 0.5$ and θ in QIII, evaluate the exact value of $\sin^6\theta - \cos^6\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q019 | WRITTEN
C05-K02-5-6-GW01

Given $\sin\theta + \cos\theta = \frac{2}{3}$ in QII, find $\sin\theta\cos\theta$ and $\sin\theta - \cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q020 | WRITTEN

C05-K02-5-6-GW02

Given $\sin\theta - \cos\theta = \frac{\sqrt{5}}{2}$ in QI, find $\sin\theta\cos\theta$ and $\sin^3\theta - \cos^3\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q021 | WRITTEN

C05-K02-5-6-GW03

Given $\sin\theta + \cos\theta = -\frac{1}{2}$ in QIII, find $\sin^3\theta + \cos^3\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q022 | WRITTEN C05-K02-5-6-GW04

Given $\sin\theta - \cos\theta = \frac{1}{3}$ in QIV, find $\sin\theta + \cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\mathrm{LHS} = \mathrm{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q023 | WRITTEN

C05-K02-5-6-GW05

Given $\sin\theta - \cos\theta = 1$ in QIII, find $\sin^3\theta - \cos^3\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q024 | MCQ

C05-K02-5-6-GM01

If $s + c = \frac{1}{2}$, then sc equals:

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q025 | MCQ

C05-K02-5-6-GM02

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

A $\frac{5}{16}$

B $\frac{11}{16}$

C $-\frac{11}{16}$

D $\frac{3}{16}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q026 | MCQ

C05-K02-5-6-GM03

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q027 | MCQ

C05-K02-5-6-GM04

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

A $\frac{1}{8}$

B $-\frac{1}{8}$

C $\frac{3}{8}$

D $-\frac{3}{8}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V03

Q028 | MCQ

C05-K02-5-6-GM05

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

A $-\frac{55\sqrt{7}}{256}$

B $\frac{55\sqrt{7}}{256}$

C $-\frac{11\sqrt{7}}{16}$

D $\frac{11}{16}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q029 | WRITTEN C05-K02-5-7-WU

Prove $\frac{\sin\theta}{1+\cos\theta} + \frac{1}{\tan\theta} = \frac{1}{\sin\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q030 | WRITTEN

C05-K02-5-7-TR

Prove (a) $\frac{1-\tan\theta}{1+\tan\theta} = \frac{\cos\theta-\sin\theta}{\cos\theta+\sin\theta}$; (b) $\frac{\cos\theta}{1-\sin\theta} - \tan\theta = \frac{1}{\cos\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Q031 | WRITTEN

C05-K02-5-7-EX1

Prove the six identities in the source Exercise: (a) $\tan^2\theta - \sin^2\theta = \tan^2\theta\sin^2\theta$; (b) $\tan\theta + 1/\tan\theta = 1/(\sin\theta\cos\theta)$; (c) $\cos\theta/(1-\sin\theta) - 1/\cos\theta = \tan\theta$; (d) $\tan\theta/\sin\theta - \sin\theta/\tan\theta = \sin\theta\tan\theta$; (e) $(1+\sin\theta)/\cos\theta + \cos\theta/(1+\sin\theta) = 2/\cos\theta$; (f) $(\tan^2\theta - \sin^2\theta)/\sin^2\theta = \tan^2\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\mathrm{LHS} = \mathrm{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V03

Q032 | WRITTEN

C05-K02-5-7-CH1

Prove that, for all defined values of x, $\frac{1+\sin x-\cos x}{1+\sin x+\cos x}+\frac{1+\sin x+\cos x}{1+\sin x-\cos x}=2\csc x$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta+\cos^2\theta=1$$

Quotient

$$\tan\theta=\frac{\sin\theta}{\cos\theta}\qquad \cot\theta=\frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1+\tan^2\theta=\frac{1}{\cos^2\theta}$$

Cubes

$$a^3+b^3=(a+b)(a^2-ab+b^2)\qquad a^3-b^3=(a-b)(a^2+ab+b^2)$$

Proof target

$$\text{LHS}=\text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q033 | WRITTEN

C05-K02-5-7-GW01

Prove $\frac{1+\sin\theta}{\cos\theta} = \frac{\cos\theta}{1-\sin\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q034 | WRITTEN C05-K02-5-7-GW02

Prove $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = 2\sec^2\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q035 | WRITTEN

C05-K02-5-7-GW03

Prove $\frac{\tan\theta}{1+\tan\theta} = \frac{\sin\theta}{\sin\theta+\cos\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Q036 | WRITTEN   C05-K02-5-7-GW04

Prove $\frac{\sin\theta}{1-\cos\theta} = \frac{1+\cos\theta}{\sin\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Q037 | WRITTEN

C05-K02-5-7-GW05

Prove $\frac{\cos\theta}{1+\sin\theta} = \frac{1-\sin\theta}{\cos\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q038 | MCQ

C05-K02-5-7-GM01

The first move in proving a trig identity is usually to:

A Change both sides at once

B Simplify one side

C Substitute random values

D Take a square root

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q039 | MCQ C05-K02-5-7-GM02

The identity used to replace $1 - \sin^2\theta$ is:

A $\cos^2\theta$

B $\tan^2\theta$

C $\sec^2\theta$

D $\sin^2\theta$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q040 | MCQ

C05-K02-5-7-GM03

$\tan\theta$ should be rewritten as:

A $\sin\theta\cos\theta$

B $\sin\theta/\cos\theta$

C $\cos\theta/\sin\theta$

D $1/\sin\theta$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Q041 | MCQ

C05-K02-5-7-GM04

To add two trig fractions, first find a:

A common numerator

B common denominator

C new quadrant

D reference angle

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q042 | MCQ

C05-K02-5-7-GM05

The result of a valid identity proof is:

A LHS=0

B LHS=RHS

C RHS=1

D undefined

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q043 | MCQ

C05-K02-5-5-GM06

If $\cos\theta = -\frac{12}{13}$ in QII, $\sin\theta$ is:

A $\frac{5}{13}$

B $-\frac{5}{13}$

C $\frac{12}{13}$

D $-\frac{12}{5}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V02

Q044 | MCQ

C05-K02-5-5-GM07

If $\tan\theta = -\frac{1}{2}$ in QII, $\cos\theta$ is:

A $\frac{2\sqrt{5}}{5}$

B $-\frac{2\sqrt{5}}{5}$

C $\frac{\sqrt{5}}{5}$

D $-\frac{\sqrt{5}}{5}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q045 | MCQ

C05-K02-5-5-GM08

If $\sin\theta = \frac{8}{17}$ in QIV, $\tan\theta$ is:

A $\frac{8}{15}$

B $-\frac{8}{15}$

C $\frac{15}{8}$

D $-\frac{15}{8}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V03

Q046 | MCQ

C05-K02-5-5-GM09

If $\tan\theta = \sqrt{3}$ and θ is in QIII, $\sin\theta$ is:

A $\frac{1}{2}$

B $-\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D $-\frac{\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q047 | MCQ

C05-K02-5-6-GM06

If $s + c = 1$, then sc equals:

A 0

B $\frac{1}{2}$

C $-\frac{1}{2}$

D 1

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q048 | MCQ

C05-K02-5-6-GM07

If $s - c = \sqrt{2}$, then sc equals:

A $-\frac{1}{2}$

B $\frac{1}{2}$

C -1

D 1

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\mathrm{LHS} = \mathrm{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q049 | MCQ

C05-K02-5-6-GM08

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

A $\frac{19}{27}$

B $\frac{5}{27}$

C $-\frac{19}{27}$

D $\frac{11}{27}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V03

Q050 | MCQ

C05-K02-5-6-GM09

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

A $\frac{11}{16}$

B $-\frac{11}{16}$

C $\frac{7}{16}$

D $-\frac{7}{16}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q051 | MCQ

C05-K02-5-7-GM06

Which identity converts $\dfrac{1}{\tan\theta}$ into sine and cosine?:

A $\dfrac{\sin\theta}{\cos\theta}$

B $\dfrac{\cos\theta}{\sin\theta}$

C $\sin\theta\cos\theta$

D $\sec\theta$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Q052 | MCQ C05-K02-5-7-GM07

The identity $1 - \cos^2\theta$ simplifies to:

A $\sin^2\theta$

B $\tan^2\theta$

C $\cos^2\theta$

D 1

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q053 | MCQ

C05-K02-5-7-GM08

When proving an identity, which side should normally be transformed?:

A Both sides simultaneously

B One side only

C Neither side

D Only the denominator

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V03

Q054 | MCQ

C05-K02-5-7-GM09

After combining trig fractions, the next step is usually to use:

A A random angle

B The Pythagorean identity

C A logarithm

D A derivative

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\mathrm{LHS} = \mathrm{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Quiz M1 | QUIZ MCQ

C05-K02-Q-M01

If $\sin\theta + \cos\theta = \frac{1}{2}$, then $\sin\theta\cos\theta$ is:

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Quiz M2 | QUIZ MCQ

C05-K02-Q-M02

If $\tan\theta = \frac{3}{4}$ in QI, then $\cos\theta$ is:

A $\frac{3}{5}$

B $\frac{4}{5}$

C $-\frac{4}{5}$

D $\frac{3}{4}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Quiz M3 | QUIZ MCQ C05-K02-Q-M03

Which identity is always valid where defined?:

A $\tan\theta = \dfrac{\cos\theta}{\sin\theta}$

B $1 + \tan^2\theta = \dfrac{1}{\cos^2\theta}$

C $\sin\theta + \cos\theta = 1$

D $\sin\theta\cos\theta = 1$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Quiz M4 | QUIZ MCQ C05-K02-Q-M04

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \quad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Quiz W1 | QUIZ WRITTEN C05-K02-Q-W01

Given $\cos\theta = -\frac{5}{13}$ in QIII, find $\sin\theta$ and $\tan\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Quiz W2 | QUIZ WRITTEN

C05-K02-Q-W02

Given $s + c = \frac{2}{3}$, find sc and $s^3 + c^3$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Quiz W3 | QUIZ WRITTEN C05-K02-Q-W03

Prove $\frac{1-\sin\theta}{\cos\theta} = \frac{\cos\theta}{1+\sin\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V02

Quiz W4 | QUIZ WRITTEN C05-K02-Q-W04

Given $\tan\theta = -2$ in QIV, find $\sin\theta$ and $\cos\theta$, then verify $1 + \tan^2\theta = 1/\cos^2\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 05

Trigonometric Functions

1 Concepts | 88 questions

B5 Landscape | one question per teaching page

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Assistant Lecturer • Faculty of Engineering • Cairo University

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Chapter 05 | Trigonometric Functions

CLASSROOM ROUTE

Concept 3 | Properties and Graphs of Trigonometric Functions

88 questions

Official lessons: 5-8, 5-9, 5-10, 5-11, 5-12

The Formula Bank changes at each Concept boundary.

Concept 3 | Properties and Graphs of Trigonometric Functions — **OFFICIAL LESSONS 5-8, 5-9, 5-10, 5-11, 5-12**

Properties and Graphs of Trigonometric Functions

Equation-first recall | use the same rail beside every question in this Concept

Parent graphs

$$y=\sin\theta,\ y=\cos\theta,\ y=\tan\theta$$

Periods

$$T_{\sin,\cos}=2\pi \quad T_{\tan}=\pi$$

Range

$$\sin,\cos:\ -1\le y\le 1 \quad \tan:\ y\in\mathbb{R}$$

Combined form

$$y=af(b\theta-q)=af\left(b\left(\theta-\frac{q}{b}\right)\right)$$

Transform order

shift $\frac{q}{b}$ $\rightarrow$ **horizontal scale** $\frac{1}{|b|}$ $\rightarrow$ **vertical scale** $|a|$

Start with the family cue, write the first move, then use the working grid.

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q001 | WRITTEN

C05-K03-5-8-WU

(1) Express $\sin\frac{9\pi}{5}$ as a trigonometric function of an acute angle. (2) Simplify $\sin\theta+\sin(\theta+\frac{\pi}{2})+\sin(\theta+\pi)+\sin(\theta+\frac{3\pi}{2})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y=\sin\theta,\ y=\cos\theta,\ y=\tan\theta$

Periods
$T_{\sin,\cos}=2\pi \quad T_{\tan}=\pi$

Range
$\sin,\cos:\ -1\le y\le 1 \quad \tan:\ y\in\mathbb{R}$

Combined form
$y=af(b\theta-q)=af\left(b\left(\theta-\frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V02

Q002 | WRITTEN

C05-K03-5-8-TR

(1) Express $\cos(-\frac{\pi}{6})$ as a trigonometric function of an acute angle. (2) Express $\sin\frac{8\pi}{7}$ as a trigonometric function of an acute angle. (3) Simplify $\cos(\frac{\pi}{2}-\theta)\sin(\pi-\theta)-\sin(\frac{3\pi}{2}+\theta)\cos(\pi-\theta)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y=\sin\theta,\ y=\cos\theta,\ y=\tan\theta$

Periods
$T_{\sin,\cos}=2\pi \quad T_{\tan}=\pi$

Range
$\sin,\cos:\ -1\le y\le 1 \quad \tan:\ y\in\mathbb{R}$

Combined form
$y=af(b\theta-q)=af\left(b\left(\theta-\frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V03

Q003 | WRITTEN

C05-K03-5-8-EX

Solve all three source demands: (1) express $\cos\frac{17\pi}{5}, \cos\frac{5\pi}{7}, \sin(-\frac{7\pi}{8}), \sin(-\frac{19\pi}{8})$ using acute angles; (2) find $\tan(-\frac{\pi}{6})$ and $\tan(-\frac{\pi}{4})$; (3) simplify (a) $\sin\theta + \cos(\pi+\theta) - \cos(\pi-\theta) + \sin(\pi+\theta)$, (b) $\sin(\theta+\frac{\pi}{2})\cos(\theta+\pi) + \sin(-\theta)\cos(\frac{\pi}{2}-\theta)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V03

Q004 | WRITTEN

C05-K03-5-8-CH

Simplify $\sin(\frac{3\pi}{2} - \theta)\sec(\pi + \theta) - \tan(\theta - \frac{\pi}{2})\cot(\frac{5\pi}{2} + \theta)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q005 | WRITTEN

C05-K03-5-9-WU

Draw the parent graphs $y = \sin\theta$, $y = \cos\theta$, and $y = \tan\theta$, and state each period and range.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

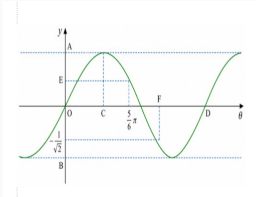
Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V02

Q006 | WRITTEN

C05-K03-5-9-WU2

For the source graph of $y = \sin\theta$, determine the labelled scales A to F.



WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

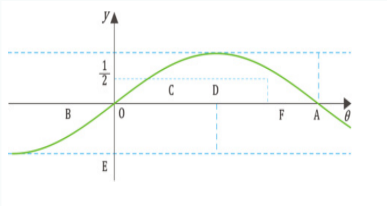
Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V02

Q007 | WRITTEN

C05-K03-5-9-TRY

(1) Draw the parent graphs $y = \sin\theta$, $y = \cos\theta$, $y = \tan\theta$, and state their periods. (2) For the source graph of $y = \cos\theta$, determine the labelled scales A to F.



WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

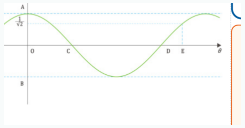
Concept 3 | Properties and Graphs of Trigonometric Functions

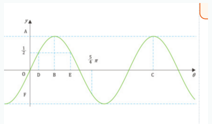
F02 | V03

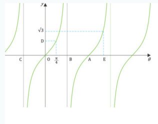
Q008 | WRITTEN

C05-K03-5-9-EX

Complete all four source graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the source cosine graph; read the labelled scales on the source sine graph; and read the labelled scales on the source tangent graph.







WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta, \ y = \cos\theta, \ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

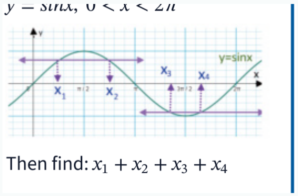
Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V04

Q009 | WRITTEN

C05-K03-5-9-CH

The source graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.



WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q010 | WRITTEN

C05-K03-5-10-WU

Draw the graphs of $y = 2\cos\theta$ and $y = \tan\frac{\theta}{2}$, and find their periods.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V02

Q011 | WRITTEN C05-K03-5-10-TRY

Draw the graphs of $y = 2\sin\theta$, $y = \tan 2\theta$, and $y = \cos\frac{\theta}{2}$, and find their periods.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V03

Q012 | WRITTEN

C05-K03-5-10-EX

Draw the nine source graphs and find each period: (a) $3\sin\theta$, (b) $3\cos\theta$, (c) $\frac{1}{2}\cos\theta$, (d) $\sin 3\theta$, (e) $\cos 2\theta$, (f) $\tan 3\theta$, (g) $\sin\frac{\theta}{2}$, (h) $\cos\frac{\theta}{3}$, (i) $\tan\frac{\theta}{3}$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V04

Q013 | WRITTEN

C05-K03-5-10-CH

Draw $y = -2|\sin(3x)|$ and find its period.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q014 | WRITTEN

C05-K03-5-11-WU

Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V02

Q015 | WRITTEN

C05-K03-5-11-TRY

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, (2) $y = \cos(\theta + \frac{\pi}{4})$, (3) $y = \tan(\theta - \frac{\pi}{2})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta, \ y = \cos\theta, \ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V03

Q016 | WRITTEN

C05-K03-5-11-EX

Draw all source graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; cosine group $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; tangent group $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V04

Q017 | WRITTEN

C05-K03-5-11-CH

Draw $y = 3\cos\left|x + \frac{\pi}{3}\right|$ and find its period.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q018 | WRITTEN

C05-K03-5-12-WU

Draw the graphs and find their periods: (1) $y = 2\cos(2\theta - \frac{\pi}{2})$, (2) $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V02

Q019 | WRITTEN

C05-K03-5-12-TRY

Draw the graphs and find their periods: (a) $y = 3\sin\frac{\theta}{2}$, (b) $y = \tan(2\theta - \frac{\pi}{2})$, (c) $y = 2\cos(2\theta - \frac{\pi}{3})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V03

Q020 | WRITTEN

C05-K03-5-12-EX

Draw all nine source graphs and find their periods: (a) $2\cos\frac{\theta}{2}$, (b) $3\sin 2\theta$, (c) $2\cos 2\theta$, (d) $2\cos(\theta-\frac{\pi}{4})$, (e) $2\sin(\theta-\frac{\pi}{3})$, (f) $\tan(2\theta-\frac{\pi}{3})$, (g) $3\cos(2\theta-\frac{\pi}{2})$, (h) $3\sin(2\theta-\frac{2\pi}{3})$, (i) $2\cos(\frac{\theta}{2}-\frac{\pi}{4})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y=\sin\theta,\ y=\cos\theta,\ y=\tan\theta$

Periods
$T_{\sin,\cos}=2\pi \quad T_{\tan}=\pi$

Range
$\sin,\cos:\ -1\le y\le 1 \quad \tan:\ y\in\mathbb{R}$

Combined form
$y=af(b\theta-q)=af\left(b\left(\theta-\frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V04

Q021 | WRITTEN

C05-K03-5-12-CH

Draw $y = 3\sin|2x - \pi|$ and find its period.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ $\to$ horizontal scale $\frac{1}{|b|}$ $\to$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q022 | MCQ

C05-K03-5-8-GM01

$\sin(\theta + \pi)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q023 | MCQ

C05-K03-5-8-GM02

$\cos(\theta + \pi)$ equals:

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q024 | MCQ

C05-K03-5-8-GM03

$\sin(\frac{\pi}{2} - \theta)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q025 | MCQ

C05-K03-5-8-GM04

$\cos(-\theta)$ equals:

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\to$ horizontal scale $\frac{1}{|b|}$ $\to$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q026 | MCQ

C05-K03-5-8-GM05

$\sin(-\theta)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q027 | MCQ

C05-K03-5-8-GM06

$\cos(\pi - \theta)$ equals:

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q028 | MCQ

C05-K03-5-8-GM07

$\sin(\frac{3\pi}{2} + \theta)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q029 | MCQ C05-K03-5-8-GM08

$\tan(\theta - \frac{\pi}{2})$ equals:

A $\tan\theta$

B $-\tan\theta$

C $\cot\theta$

D $-\cot\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q030 | MCQ C05-K03-5-9-GM01

The period of $y = \sin\theta$ is:

A $\pi/2$

B π

C 2π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q031 | MCQ

C05-K03-5-9-GM02

The range of $y = \cos\theta$ is:

A $y \geq 0$

B $-1 \leq y \leq 1$

C $y \in \mathbb{R}$

D y>1

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \leq y \leq 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q032 | MCQ

C05-K03-5-9-GM03

The period of $y = \tan\theta$ is:

A $\pi/2$

B π

C 2π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q033 | MCQ C05-K03-5-9-GM04

A vertical asymptote of $y = \tan\theta$ occurs at:

A $\theta = k\pi$

B $\theta = \dfrac{\pi}{2} + k\pi$

C $\theta = 2k\pi$

D $\theta = \dfrac{\pi}{4} + k\pi$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta, \ y = \cos\theta, \ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q034 | MCQ

C05-K03-5-9-GM05

The maximum value of $\sin\theta$ is:

A -1

B 0

C 1

D π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q035 | MCQ C05-K03-5-9-GM06

The x-coordinate of the first positive sine maximum is:

A $\pi/4$

B $\pi/2$

C π

D $3\pi/2$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q036 | MCQ

C05-K03-5-9-GM07

At $\theta = 0$, $\cos\theta$ equals:

A -1

B 0

C 1

D undefined

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q037 | MCQ

C05-K03-5-9-GM08

At $\theta = 0$, $\tan\theta$ equals:

A -1

B 0

C 1

D undefined

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \leq y \leq 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q038 | MCQ C05-K03-5-10-GM01

The range of $y = 3\sin\theta$ is:

A [-1,1]

B [-3,3]

C [0,3]

D $\mathbb{R}$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q039 | MCQ

C05-K03-5-10-GM02

The period of $y = \sin 4\theta$ is:

A $\pi/4$

B $\pi/2$

C π

D 2π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q040 | MCQ

C05-K03-5-10-GM03

The period of $y = \cos(\theta/3)$ is:

A $2\pi/3$

B 2π

C 3π

D 6π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q041 | MCQ

C05-K03-5-10-GM04

The period of $y = \tan 5\theta$ is:

A $\pi/5$

B π

C 5π

D $2\pi/5$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q042 | MCQ

C05-K03-5-10-GM05

The range of $y = \frac{1}{2}\cos\theta$ is:

A [-2,2]

B [-1,1]

C [-1/2,1/2]

D [0,1/2]

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q043 | MCQ

C05-K03-5-10-GM06

The graph of $y = -2\sin\theta$ is the parent sine graph:

A shifted only

B reflected and stretched vertically

C compressed horizontally only

D undefined

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q044 | MCQ

C05-K03-5-10-GM07

The period of $y = \tan(\theta/2)$ is:

A $\pi/2$

B π

C 2π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q045 | MCQ

C05-K03-5-10-GM08

The period of $y = \sin(3\theta)$ is:

A $\pi/3$

B $2\pi/3$

C 3π

D 6π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q046 | MCQ

C05-K03-5-11-GM01

The graph of $y = \sin(\theta - \pi/4)$ is shifted:

A left by $\pi/4$

B right by $\pi/4$

C up by $\pi/4$

D down by $\pi/4$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q047 | MCQ

C05-K03-5-11-GM02

The graph of $y = \cos(\theta + \pi/3)$ is shifted:

A right by $\pi/3$

B left by $\pi/3$

C up by $\pi/3$

D down by $\pi/3$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q048 | MCQ

C05-K03-5-11-GM03

A horizontal shift changes the sine period from 2π to:

A π

B 2π

C 4π

D 0

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q049 | MCQ

C05-K03-5-11-GM04

A horizontal shift changes the tangent period from π to:

A $\pi/2$

B π

C 2π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q050 | MCQ

C05-K03-5-11-GM05

The graph of $y = \sin(\theta + \pi/2)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q051 | MCQ

C05-K03-5-11-GM06

The graph of $y = \cos(\theta - \pi)$ equals:

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\to$ horizontal scale $\frac{1}{|b|}$ $\to$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q052 | MCQ

C05-K03-5-11-GM07

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:

A $\theta = 0$

B $\theta = \pi/2$

C $\theta = \pi$

D $\theta = 2\pi$

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q053 | MCQ

C05-K03-5-11-GM08

Since cosine is even, $\cos|x|$ equals:

A $\cos x$

B $-\cos x$

C $\sin x$

D $|\cos x|$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q054 | MCQ C05-K03-5-12-GM01

In $y = a\sin(b\theta - q)$, the horizontal shift is:

A q

B q/b

C b/q

D a/b

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q055 | MCQ

C05-K03-5-12-GM02

The period of $y = 2\cos(3\theta - \pi)$ is:

A $\pi/3$

B $2\pi/3$

C π

D 2π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q056 | MCQ C05-K03-5-12-GM03

The range of $y = 4\sin(2\theta - 1)$ is:

A [-1,1]

B [-2,2]

C [-4,4]

D $\mathbb{R}$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q057 | MCQ C05-K03-5-12-GM04

For $y = \tan(2\theta - \pi/3)$, the shift is:

A left by $\pi/3$

B right by $\pi/6$

C right by $\pi/3$

D left by $\pi/6$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta, \ y = \cos\theta, \ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q058 | MCQ

C05-K03-5-12-GM05

The period of $y = 3\sin(\theta/2)$ is:

A π

B 2π

C 3π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q059 | MCQ C05-K03-5-12-GM06

The correct transformation order for $y = af(b\theta - q)$ starts with:

A vertical scale

B horizontal shift

C horizontal scale

D reflection only

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q060 | MCQ

C05-K03-5-12-GM07

The period of $y = \tan(4\theta + q)$ is:

A $\pi/4$

B $\pi/2$

C π

D 2π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q061 | MCQ

C05-K03-5-12-GM08

The range of $y = 3\sin|2x - \pi|$ is:

A [-3,3]

B [0,3]

C [-1,1]

D $\mathbb{R}$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q062 | WRITTEN

C05-K03-5-8-GW01

Express $\cos \dfrac{11\pi}{6}$ as a function of an acute angle.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q063 | WRITTEN C05-K03-5-8-GW02

Simplify $\sin(\theta + \pi) - \cos(\frac{\pi}{2} - \theta)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q064 | WRITTEN

C05-K03-5-8-GW03

Simplify $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q065 | WRITTEN

C05-K03-5-8-GW04

Evaluate $\tan(-\frac{3\pi}{4})$ using an acute reference angle.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q066 | WRITTEN

C05-K03-5-9-GW01

State the period and range of $y = \cos\theta$, then list its key points on one cycle.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q067 | WRITTEN

C05-K03-5-9-GW02

State the period and asymptotes of $y = \tan\theta$ on $[-\pi, \pi]$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q068 | WRITTEN

C05-K03-5-9-GW03

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q069 | WRITTEN

C05-K03-5-9-GW04

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta, \ y = \cos\theta, \ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q070 | WRITTEN

C05-K03-5-10-GW01

Find the period and range of $y = 4\cos\theta$.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q071 | WRITTEN

C05-K03-5-10-GW02

Find the period and range of $y = \sin(5\theta)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\to$ horizontal scale $\frac{1}{|b|}$ $\to$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q072 | WRITTEN

C05-K03-5-10-GW03

Find the period and range of $y = \tan(\theta/4)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q073 | WRITTEN

C05-K03-5-10-GW04

Describe the transformations from $y = \cos\theta$ to $y = -\frac{1}{2}\cos(2\theta)$.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q074 | WRITTEN C05-K03-5-11-GW01

Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q075 | WRITTEN

C05-K03-5-11-GW02

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q076 | WRITTEN

C05-K03-5-11-GW03

Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q077 | WRITTEN

C05-K03-5-12-GW01

For $y = 2\sin(4\theta - \pi)$, state the shift, period and range.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\to$ horizontal scale $\frac{1}{|b|}$ $\to$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q078 | WRITTEN

C05-K03-5-12-GW02

For $y = \cos(\theta/3 + \pi/6)$, state the shift and period.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q079 | WRITTEN

C05-K03-5-12-GW03

For $y = 3\tan(2\theta + \pi)$, state the shift and period.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q080 | WRITTEN

C05-K03-5-12-GW04

For $y = 2\cos(\theta/2 - \pi/8)$, state the shift, period and range.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Quiz M1 | QUIZ MCQ

C05-K03-Q-M01

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:

A $\pi/2$

B π

C 2π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Quiz M2 | QUIZ MCQ

C05-K03-Q-M02

$\sin(\theta + \pi)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Quiz M3 | QUIZ MCQ

C05-K03-Q-M03

The range of $y = 2\cos\theta$ is:

A [-1,1]

B [-2,2]

C [0,2]

D $\mathbb{R}$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Quiz M4 | QUIZ MCQ

C05-K03-Q-M04

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

A $left by \frac{\pi}{4}$

B $right by \frac{\pi}{4}$

C $up by \frac{\pi}{4}$

D $down by \frac{\pi}{4}$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Quiz W1 | QUIZ WRITTEN — C05-K03-Q-W01

For $y = 3\sin(2\theta - \pi)$, state the horizontal shift, period and range.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Quiz W2 | QUIZ WRITTEN

C05-K03-Q-W02

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V02

Quiz W3 | QUIZ WRITTEN C05-K03-Q-W03

Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V04

Quiz W4 | QUIZ WRITTEN

C05-K03-Q-W04

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 05

Trigonometric Functions

1 Concepts | 42 synchronized questions

B5 Landscape | one question per teaching page

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Chapter 05 | Trigonometric Functions

CLASSROOM ROUTE

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

42 questions

Official lessons: 5-13, 5-14, 5-15, 5-16

The Formula Bank changes automatically at each Concept boundary. Every question ID and order remains synchronized with the approved A4 Classified and Solutions pair.

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions
OFFICIAL LESSONS 5-13, 5-14, 5-15, 5-16

Solving Equations and Inequalities Involving Trigonometric Functions

Equation-first recall | use the same rail beside every question in this Concept

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Start with the family cue, write the first move, then use the working grid.

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Q001 | WRITTEN

C05-K04-5-13-WU

Solve (a) $-2\cos\theta = \sqrt{2}$ and (b) $\tan\theta + \sqrt{3} = 0$, giving the general solution for (b).

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V02

Q002 | WRITTEN

C05-K04-5-13-TRY

Solve (a) $2\cos\theta - 1 = 0$ and (b) $\tan\theta - 1 = 0$ for $0 \le \theta \le 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$\sin^2\theta + \cos^2\theta = 1$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range check

$-1 \le \sin\theta, \cos\theta \le 1$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V03

Q003 | WRITTEN

C05-K04-5-13-EX

Solve all source equations on $0 \le \theta \le 2\pi$: (a) $2\sin\theta + 1 = 0$, (b) $2\sin\theta = \sqrt{3}$, (c) $3\sin\theta + 3 = 0$, (d) $\sqrt{2}\cos\theta + 1 = 0$, (e) $2\cos\theta = \sqrt{3}$, (f) $2\cos\theta - \sqrt{2} = 0$, (g) $\tan\theta + 1 = 0$, (h) $\sqrt{3}\tan\theta = 1$, (i) $\sqrt{3}\tan\theta = 3$, and give general solutions for (j) $2\sin\theta + \sqrt{3} = 0$, (k) $\cos\theta + 1 = 0$, (l) $\sqrt{3}\tan\theta + 3 = 0$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V04

Q004 | WRITTEN C05-K04-5-13-CH

Solve $\tan^2\theta - (2+\sqrt{3})\tan\theta + 2\sqrt{3} = 0$ and give the general solution.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Q005 | WRITTEN

C05-K04-5-14-WU

Solve for $0 \le \theta < 2\pi$: (a) $2\cos\theta > -\sqrt{3}$, (b) $\tan\theta \le \frac{1}{\sqrt{3}}$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V02

Q006 | WRITTEN

C05-K04-5-14-TRY

Solve for $0 \le \theta < 2\pi$: (a) $2\sin\theta \le -1$, (b) $\cos\theta > \frac{1}{2}$, (c) $\tan\theta > 1$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V03

Q007 | WRITTEN

C05-K04-5-14-EX

Solve all source inequalities for $0 \le \theta < 2\pi$: (a) $\sin\theta > \frac{\sqrt{3}}{2}$, (b) $\sin\theta - \frac{\sqrt{2}}{2} > 0$, (c) $2\sin\theta + \sqrt{3} \le 0$, (d) $\cos\theta > 0$, (e) $\cos\theta - \frac{1}{2} \le 0$, (f) $2\cos\theta + \sqrt{2} \le 0$, (g) $\tan\theta \le -\frac{1}{\sqrt{3}}$, (h) $\tan\theta + \sqrt{3} > 0$, (i) $\tan\theta + 1 \ge 0$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V04

Q008 | WRITTEN

C05-K04-5-14-CH

Solve $2\sin^2\theta - \cos\theta - 1 > 0$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Q009 | WRITTEN

C05-K04-5-15-WU

Solve for $0 \le \theta < 2\pi$: (a) $2\cos^2\theta - 3\sin\theta = 0$, (b) $2\sin^2\theta - 3\cos\theta - 3 \le 0$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V02

Q010 | WRITTEN

C05-K04-5-15-TRY

Solve for $0 \le \theta < 2\pi$: (a) $2\sin^2\theta - \sin\theta = 0$, (b) $2\cos^2\theta - \sin\theta = 2$, (c) $2\sin^2\theta - 4 < 5\cos\theta$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V03

Q011 | WRITTEN　　C05-K04-5-15-EX

Solve all source equations and inequalities for $0 \le \theta < 2\pi$: (a) $2\sin^2\theta + \sin\theta - 1 = 0$, (b) $2\sin^2\theta + 3\cos\theta = 0$, (c) $2\sin^2\theta - \cos\theta = 2$, (d) $2\cos^2\theta - 5\cos\theta - 3 = 0$, (e) $2\cos^2\theta + \sin\theta - 1 = 0$, (f) $2\cos^2\theta + 3\sin\theta = 3$, (g) $2\cos^2\theta + 3\sin\theta \le 0$, (h) $2\sin^2\theta + \cos\theta > 1$, (i) $2\cos^2\theta - 4 \le 5\sin\theta$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$　find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V04

Q012 | WRITTEN

C05-K04-5-15-CH

Solve $\frac{1-2\sin^2\theta+\sin\theta}{\cos^2\theta} \geq 1$ for $0 \leq \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$\sin^2\theta + \cos^2\theta = 1$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range check

$-1 \leq \sin\theta, \cos\theta \leq 1$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q013 | WRITTEN

C05-K04-5-16-WU

Solve for $0 \le \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$, (b) $\sin(\theta + \frac{\pi}{6}) > -\frac{1}{2}$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V02

Q014 | WRITTEN

C05-K04-5-16-TRY

Solve for $0 \le \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{6}) = \frac{1}{2}$, (b) $\tan(\theta + \frac{\pi}{4}) = \sqrt{3}$, (c) $\cos(\theta + \frac{\pi}{4}) < -\frac{\sqrt{2}}{2}$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V03

Q015 | WRITTEN

C05-K04-5-16-EX

Solve for $0 \le \theta < 2\pi$: (a) $\cos(\theta + \frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$, (b) $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$, (c) $\sin(\theta + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$, (d) $\tan(\theta + \frac{\pi}{4}) = -1$, (e) $\tan(\theta + \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$, (f) $\tan(\theta + \frac{\pi}{6}) = 1$, (g) $\sin(\theta + \frac{\pi}{6}) \le \frac{\sqrt{3}}{2}$, (h) $\sin(\theta + \frac{\pi}{4}) \ge \frac{\sqrt{2}}{2}$, (i) $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V04

Q016 | WRITTEN

C05-K04-5-16-CH

Solve $4\sin^2(\theta + \frac{\pi}{6}) \geq 1$ for $0 \leq \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V04

Q017 | WRITTEN

C05-K04-5-16-REF

A student divides $\tan x \times \cos x = \frac{1}{2}$ by $\cos x$, obtaining $\tan x = \frac{1}{2\cos x}$. Evaluate this strategy and propose a better substitution on $[0°, 360°[$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Q018 | MCQ

C05-K04-5-13-GM01

The period used for a general tangent solution is:

A 2π

B π

C $\pi/2$

D 4π

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Q019 | MCQ C05-K04-5-13-GM02

The solutions of $2\cos\theta = 1$ on $[0, 2\pi]$ are:

A $\pi/6, 11\pi/6$

B $\pi/3, 5\pi/3$

C $2\pi/3, 4\pi/3$

D $\pi/4, 7\pi/4$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Q020 | MCQ

C05-K04-5-13-GM03

Which root is impossible when solving $2\sin^2\theta + 3\sin\theta - 2 = 0$?:

A $\sin\theta = 1/2$

B $\sin\theta = -2$

C $\sin\theta = 0$

D $\sin\theta = -1/2$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Q021 | MCQ

C05-K04-5-13-GM04

The quadratic challenge factors into roots:

A 1 and $\sqrt{3}$

B 2 and $\sqrt{3}$

C 2 and 3

D $\pi/3$ and 2

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Q022 | MCQ

C05-K04-5-14-GM01

For $2\cos\theta > -\sqrt{3}$, the excluded arc is:

A $[0, 5\pi/6]$

B $[5\pi/6, 7\pi/6]$

C $[7\pi/6, 2\pi)$

D $(\pi/6, 5\pi/6)$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Q023 | MCQ · C05-K04-5-14-GM02

The inequality $\tan\theta > 1$ on $[0, 2\pi)$ holds on:

A $(\pi/4, \pi/2) \cup (5\pi/4, 3\pi/2)$

B $[0, \pi/4]$

C $(\pi/2, \pi)$

D $[\pi, 2\pi)$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Q024 | MCQ

C05-K04-5-14-GM03

A closed endpoint in a trig inequality is used when the sign is:

A strictly greater

B strictly less

C greater than or equal to

D never

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Q025 | MCQ

C05-K04-5-14-GM04

The 5-14 challenge reduces to which cosine condition?:

A $\cos\theta > 1/2$

B $-1 < \cos\theta < 1/2$

C $\cos\theta \le -1$

D $\cos\theta \ge 1$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Q026 | MCQ

C05-K04-5-15-GM01

When a quadratic gives $\sin\theta = -2$, we:

A keep it

B reject it

C square it

D divide by 2

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Q027 | MCQ

C05-K04-5-15-GM02

The first step for $2\cos^2\theta - 3\sin\theta = 0$ is to use:

A $\sin^2\theta + \cos^2\theta = 1$

B $\tan\theta = \sin\theta$

C $\cos\theta = 1$

D the tangent period

WORKING AREA

FORMULA BANK

Single-function form
$\sin^2\theta + \cos^2\theta = 1$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range check
$-1 \le \sin\theta, \cos\theta \le 1$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Q028 | MCQ

C05-K04-5-15-GM03

The solution of $2\sin^2\theta + \sin\theta - 1 = 0$ includes:

A $\theta = \pi/6$

B $\theta = \pi/4$

C $\theta = \pi/3$

D $\theta = 2\pi/3$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Q029 | MCQ

C05-K04-5-15-GM04

In the rational challenge, which angle must be excluded?:

A 0

B $\pi/4$

C $\pi/2$

D π

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q030 | MCQ

C05-K04-5-16-GM01

After setting $t = \theta + \pi/3$ with $0 \le \theta < 2\pi$, the t-range is:

A $[0, 2\pi)$

B $[\pi/3, 7\pi/3)$

C $[-\pi/3, 5\pi/3)$

D $[\pi, 3\pi)$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q031 | MCQ

C05-K04-5-16-GM02

The solutions of $\tan(\theta + \pi/4) = \sqrt{3}$ include:

A $\pi/12$

B $\pi/6$

C $\pi/3$

D $\pi/2$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q032 | MCQ

C05-K04-5-16-GM03

For $\sin(\theta + \pi/4) \geq \sqrt{2}/2$, the θ-solution is:

A $[0, \pi/2]$

B $(0, \pi/2)$

C $[\pi/2, \pi]$

D $[0, 2\pi)$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q033 | MCQ

C05-K04-5-16-GM04

The 5-16 challenge is equivalent to:

A $|\sin t| \geq 1/2$

B $\sin t = 0$

C $\cos t > 1/2$

D $\tan t = 1$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q034 | MCQ C05-K04-5-16-GM05

In the Reflect problem, $\tan x \cos x$ simplifies to:

A $\cos x$

B $\sin x$

C $\tan^2 x$

D 1

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Quiz M1 | QUIZ MCQ

C05-K04-Q-M01

The general solution of $\tan\theta = -1$ is:

A $-\pi/4 + 2n\pi$

B $-\pi/4 + n\pi$

C $\pi/4 + 2n\pi$

D $\pi/2 + n\pi$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Quiz M2 | QUIZ MCQ

C05-K04-Q-M02

The solution of $\cos\theta > 0$ on $[0, 2\pi)$ is:

A $(\pi/2, 3\pi/2)$

B $[0, \pi/2) \cup (3\pi/2, 2\pi)$

C $[0, \pi)$

D $(\pi, 2\pi)$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Quiz M3 | QUIZ MCQ

C05-K04-Q-M03

The valid sine root of $2\sin^2\theta + 3\sin\theta - 2 = 0$ is:

A $\sin\theta = -2$

B $\sin\theta = 1/2$

C $\sin\theta = 2$

D $\sin\theta = -1/2$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Quiz M4 | QUIZ MCQ

C05-K04-Q-M04

The first operation in a shifted-argument problem is to:

A divide by $\cos\theta$

B set t equal to the whole argument and find its range

C change radians to degrees

D square both sides

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ **find the range of** t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Quiz W1 | QUIZ WRITTEN C05-K04-Q-W01

Solve $2\cos\theta = \sqrt{3}$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Quiz W2 | QUIZ WRITTEN

C05-K04-Q-W02

Solve $\tan\theta \le -1$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Quiz W3 | QUIZ WRITTEN

C05-K04-Q-W03

Solve $2\sin^2\theta + \sin\theta - 1 = 0$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$\sin^2\theta + \cos^2\theta = 1$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range check

$-1 \le \sin\theta, \cos\theta \le 1$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Quiz W4 | QUIZ WRITTEN

C05-K04-Q-W04

Solve $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$ for $0 \leq \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 05

Trigonometric Functions

1 Concepts | 62 questions

B5 Landscape | one question per teaching page

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Chapter 05 | Trigonometric Functions

CLASSROOM ROUTE

Concept 5 | Advanced Theorems and Applications

62 questions

Official lessons: 5-17, 5-18, 5-19, 5-20, 5-21, 5-22, 5-23, 5-24, 5-25

The Formula Bank changes at each Concept boundary.

Concept 5 | Advanced Theorems and Applications — OFFICIAL LESSONS 5-17, 5-18, 5-19, 5-20, 5-21, 5-22, 5-23, 5-24, 5-25

Advanced Theorems and Applications

Equation-first recall | use the same rail beside every question in this Concept

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Start with the family cue, write the first move, then use the working grid.

Concept 5 | Advanced Theorems and Applications

F01 | V01

Q001 | WRITTEN

C05-K05-5-17-WU

Find the maximum and minimum of $y = sin^2\theta + cos\theta + 1,\ 0le\theta < 2pi$, and the corresponding θ.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | V02

Q002 | WRITTEN

C05-K05-5-17-TRY

Find the maximum and minimum of $y = cos^2\theta - sin\theta, 0le\theta < 2pi$, and the corresponding θ.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | V03

Q003 | WRITTEN

C05-K05-5-17-EX

Solve the four source maximum/minimum problems on $0 le \theta < 2 pi$, including $y = \sin^2\theta + \sin\theta - 1$, $y = \cos^2\theta - \cos\theta + 1$, $y = 2\sin\theta + \cos^2\theta - 1$, and $y = -2\sin^2\theta - 2\cos\theta + 1$. Include the source challenge $y = \cos^2\theta - \sin\theta$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | V01

Q004 | WRITTEN

C05-K05-5-18-WU

Use addition theorems to find $\sin 75°$ and $\cos(\pi/12)$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | V02

Q005 | WRITTEN

C05-K05-5-18-TRY

Given $\sin\alpha = \frac{\sqrt{3}}{2}$ in QII and $\cos\beta = -\frac{3}{5}$ in QIII, find $\sin(\alpha+\beta)$ and $\cos(\alpha-\beta)$.

WORKING AREA

FORMULA BANK

Extrema

$y = -(t-h)^2 + k$

Addition

$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

Double angle

$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$

Harmonic form

$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$

Angle / range

$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$

Concept 5 | Advanced Theorems and Applications

F02 | V03

Q006 | WRITTEN

C05-K05-5-18-EX

Complete the source addition-theorem exercises $exactvalues,\ quadrant-controlled\alpha,\ \beta,\ and the three-angle challenge(\sin\alpha = 0.6,\ \cos\beta = 0.8,\ \tan\gamma = 0.75)$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | V01

Q007 | WRITTEN

C05-K05-5-19-WU

Find $\tan 105°$, then find the angle between $y = 2x + 1$ and $y = -3x + 2$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | V02

Q008 | WRITTEN C05-K05-5-19-TRY

Find the angle between each source pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x + 2y - 3 = 0$ with $-5x + 3y + 1 = 0$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | V03

Q009 | WRITTEN C05-K05-5-19-EX

Find the exact angles for all source line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan\theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y-axis.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | V01

Q010 | WRITTEN C05-K05-5-20-WU

Given $\sin\alpha = 1/3$ with α in QII, find $\sin 2\alpha,\ \cos 2\alpha,\ \tan 2\alpha$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | V02

Q011 | WRITTEN

C05-K05-5-20-TRY

Given $\sin\alpha = -2/3$ with $2\pi < \alpha < 3\pi$, find $\sin 2\alpha,\ \cos 2\alpha,\ \tan 2\alpha$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | V03

Q012 | WRITTEN C05-K05-5-20-EX

Solve the source double-angle exercises for given sine/cosine values and the challenge $\cos\theta = 0.6,\ \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | V01

Q013 | WRITTEN

C05-K05-5-21-WU

Given $\cos\alpha = 3/5, 0 < \alpha < \pi$, find $\sin(\alpha/2),\ \cos(\alpha/2),\ \tan(\alpha/2)$, and evaluate $\cos(5\pi/8)$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | V02

Q014 | WRITTEN

C05-K05-5-21-TRY

Given $\cos\alpha = -1/3$, $0 < \alpha < \pi$, find $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, and evaluate $\sin(3\pi/8)$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | V03

Q015 | WRITTEN

C05-K05-5-21-EX

Solve the source half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}$, $4\pi < A < 5\pi$, $\cos B = -\frac{\sqrt{5}}{3}$, $5\pi < B < 6\pi$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | V01

Q016 | WRITTEN

C05-K05-5-22-WU

Solve on $0 \le \theta < 2\pi$: $a\sin 2\theta + \sin\theta = 0$, $b\cos 2\theta - 7\cos\theta + 4 \ge 0$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | V02

Q017 | WRITTEN

C05-K05-5-22-TRY

Solve the source equation $\sin 2\theta + \cos\theta = 0$ on $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | V03

Q018 | WRITTEN

C05-K05-5-22-EX

Solve all source double-angle equations and inequalities on $0 \leq \theta < 2\pi$, including the challenge $\cos 2\theta - 7\cos\theta + 4 \geq 0$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \leq \sin\theta, \cos\theta \leq 1$$

Concept 5 | Advanced Theorems and Applications

F07 | V01

Q019 | WRITTEN

C05-K05-5-23-WU

Rewrite $\sin\theta - \sqrt{3}\cos\theta$ as $r\sin(\theta+\alpha)$, $r>0$, $-\pi<\alpha<\pi$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F07 | V02

Q020 | WRITTEN C05-K05-5-23-TRY

Rewrite the source expressions $\sin\theta - \cos\theta$, $\sin\theta + \cos\theta$, $3\sin\theta - \sqrt{3}\cos\theta$, and $3\sin\theta - \cos\theta$ in harmonic-addition form.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F07 | V03

Q021 | WRITTEN

C05-K05-5-23-EX

Rewrite all source harmonic-addition exercises in the form $r\sin(\theta+\alpha)$, including $2\sin\theta + 2\cos\theta$ and the challenge $\sin\theta - \sqrt{3}\cos\theta$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | V01

Q022 | WRITTEN

C05-K05-5-24-WU

Solve on $0 \le \theta < 2\pi$: $a \sin\theta - \cos\theta = 1$, $b \sin\theta + \cos\theta < 3/2$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | V02

Q023 | WRITTEN

C05-K05-5-24-TRY

Solve $\sin\theta \le \cos\theta$ on $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | V03

Q024 | WRITTEN

C05-K05-5-24-EX

Solve all source harmonic-addition equations and inequalities on $0 \le \theta < 2\pi$, including $\sin\theta + \cos\theta + 1 = 0$ and $\sin\theta - \cos\theta + 3/2 > 0$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | V01

Q025 | WRITTEN

C05-K05-5-25-WU

Find the maximum and minimum of $y = 2\sin(\theta - \pi/3)$, and of $y = 2\cos 2\theta + 4\cos\theta + 1$, $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | V02

Q026 | WRITTEN

C05-K05-5-25-TRY

Find the maximum and minimum of the source functions $y = -\frac{4}{3}\sin^2\theta + \cos\theta$ and $y = \cos 2\theta - 2\cos\theta$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | V03

Q027 | WRITTEN

C05-K05-5-25-EX

Solve all source maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | C05-K05-F01-VMCQ

Q028 | MCQ

C05-K05-5-17-GM01

The substitution for $\sin^2\theta + \cos\theta + 1$ should be Which of the following is correct?:

A $t = sintheta$

B $t = costheta$

C $t = tantheta$

D $t = theta$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | C05-K05-F01-VMCQ

Q029 | MCQ C05-K05-5-17-GM02

The maximum of $\sin^2\theta + \cos\theta + 1$ is:

A 0	**B** 1
C 9/4	**D** 4

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | C05-K05-F01-VMCQ

Q030 | MCQ

C05-K05-5-17-GM03

An extremal vertex is valid only if t lies in Which of the following is correct?:

A $[-2, 2]$

B $[-1, 1]$

C [0,1]

D all real numbers

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

Double angle
$\sin 2\alpha = 2\sin\alpha\cos\alpha$ $\quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$

Harmonic form
$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$

Angle / range
$\tan\theta = \dfrac{m_1 - m_2}{1 + m_1 m_2}$ $\quad -1 \le \sin\theta, \cos\theta \le 1$

Concept 5 | Advanced Theorems and Applications

F02 | C05-K05-F02-VMCQ

Q031 | MCQ

C05-K05-5-18-GM04

The identity for sin(a+b) is:

A $sinalphacosbeta + cosalphasinbeta$

B sin alpha sin beta

C cos alpha cos beta

D $tanalpha + tanbeta$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | C05-K05-F02-VMCQ

Q032 | MCQ

C05-K05-5-18-GM05

sin 75 degrees equals Which of the following is correct?:

A $(sqrt6 - sqrt2)/4$

B $(sqrt6 + sqrt2)/4$

C $1/2$

D $sqrt3/2$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t - h)^2 + k$$

Addition
$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta + \alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | C05-K05-F02-VMCQ

Q033 | MCQ
C05-K05-5-18-GM06

A missing trig value is selected using:

A the quadrant

B the denominator

C the period only

D a graph scale

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | C05-K05-F03-VMCQ

Q034 | MCQ

C05-K05-5-19-GM07

The slope of y=mx+k is:

A k

B m

C $1/m$

D $m+k$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | C05-K05-F03-VMCQ

Q035 | MCQ

C05-K05-5-19-GM08

The angle formula uses denominator Which of the following is correct?:

A $1 + m1m2$

B $m1 - m2$

C $m1 + m2$

D m1m2

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t - h)^2 + k$$

Addition
$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta + \alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | C05-K05-F03-VMCQ

Q036 | MCQ

C05-K05-5-19-GM09

For an acute angle, if tan theta1 is negative we use Which of the following is correct?:

A theta1

B $pi - theta1$

C 2theta1

D $-theta1only$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | C05-K05-F04-VMCQ

Q037 | MCQ

C05-K05-5-20-GM10

sin(2a) equals Which of the following is correct?:

A sin alpha

B 2 sin alpha cos alpha

C cos squared alpha

D tan alpha

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | C05-K05-F04-VMCQ

Q038 | MCQ C05-K05-5-20-GM11

cos(2a) can be written as Which of the following is correct?:

A $cossquaredalpha - sinsquaredalpha$

B 2sin alpha

C tan alpha

D $1 + sinalpha$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | C05-K05-F04-VMCQ

Q039 | MCQ

C05-K05-5-20-GM12

For theta in QIV, sin theta is:

A positive

B negative

C zero always

D undefined

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | C05-K05-F05-VMCQ

Q040 | MCQ

C05-K05-5-21-GM13

The half-angle formula for sin(a/2) uses Which of the following is correct?:

A $(1 - cosalpha)/2$

B $(1 + cosalpha)/2$

C tan alpha

D cos 2alpha

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | C05-K05-F05-VMCQ

Q041 | MCQ

C05-K05-5-21-GM14

The sign of a half-angle root is chosen from Which of the following is correct?:

A the original quadrant interval

B the coefficient only

C the denominator

D the period of tan

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | C05-K05-F05-VMCQ

Q042 | MCQ C05-K05-5-21-GM15

If cos a=-1/3 and 0<a<pi, cos(a/2) is:

A negative

B positive

C zero

D undefined

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | C05-K05-F06-VMCQ

Q043 | MCQ C05-K05-5-22-GM16

For double-angle equations, first rewrite in Which of the following is correct?:

A one trig function

B two unrelated angles

C degrees only

D a logarithm

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | C05-K05-F06-VMCQ

Q044 | MCQ

C05-K05-5-22-GM17

The factorisation of sin(2 theta)+sin theta is:

A $sintheta(2costheta + 1)$

B $costheta(2sintheta - 1)$

C $sintheta + costheta$

D 2sin theta

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t - h)^2 + k$$

Addition
$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta + \alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | C05-K05-F06-VMCQ

Q045 | MCQ

C05-K05-5-22-GM18

A tangent asymptote is always Which of the following is correct?:

A included

B excluded

C a maximum

D an endpoint

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F07 | C05-K05-F07-VMCQ

Q046 | MCQ

C05-K05-5-23-GM19

For a sin theta+b cos theta, r is:

A $a+b$

B $sqrt(a^2+b^2)$

C $a-b$

D ab

WORKING AREA

FORMULA BANK

Extrema
$$y=-(t-h)^2+k$$

Addition
$$\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha=2\sin\alpha\cos\alpha \qquad \cos 2\alpha=\cos^2\alpha-\sin^2\alpha$$

Harmonic form
$$a\sin\theta+b\cos\theta=r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta=\frac{m_1-m_2}{1+m_1m_2} \qquad -1\le\sin\theta,\cos\theta\le 1$$

Concept 5 | Advanced Theorems and Applications

F07 | C05-K05-F07-VMCQ

Q047 | MCQ

C05-K05-5-23-GM20

The harmonic target form is:

A $rsin(theta + alpha)$

B r cos theta

C a tan theta

D sin theta only

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F07 | C05-K05-F07-VMCQ

Q048 | MCQ

C05-K05-5-23-GM21

The angle a is located from point Which of the following is correct?:

A P(a,b)

B P(r,theta)

C the origin only

D the x intercept

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | C05-K05-F08-VMCQ

Q049 | MCQ

C05-K05-5-24-GM22

After synthesis, t is usually Which of the following is correct?:

A $theta + alpha$

B $theta^2$

C tan theta

D cos theta

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | C05-K05-F08-VMCQ

Q050 | MCQ

C05-K05-5-24-GM23

If max(sin theta+cos theta)=sqrt2, then sin theta+cos theta<3/2 is:

A never true

B always true

C true only at pi

D undefined

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | C05-K05-F08-VMCQ

Q051 | MCQ

C05-K05-5-24-GM24

In a strict inequality, equality endpoints are:

A included

B excluded

C doubled

D random

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | C05-K05-F09-VMCQ

Q052 | MCQ

C05-K05-5-25-GM25

The range of 2sin(theta-pi/3) is:

A $[-1, 1]$

B $[-2, 2]$

C [0,2]

D $[-3, 3]$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | C05-K05-F09-VMCQ

Q053 | MCQ C05-K05-5-25-GM26

For a double-angle maximum problem, use Which of the following is correct?:

A $thedouble - angleformulathenaboundedvariable$

B a logarithm

C only a graph

D no substitution

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | C05-K05-F09-VMCQ

Q054 | MCQ

C05-K05-5-25-GM27

A vertex outside [-1,1] is:

A an allowed extremum

B rejected

C always the maximum

D always the minimum

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | C05-K05-F09-VQUIZ

Quiz M1 | QUIZ MCQ

C05-K05-Q-M01

The maximum of $2\sin(\theta - \pi/3)$ is:

A -2	**B** -1
C 1	**D** 2

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | C05-K05-F02-VQUIZ

Quiz M2 | QUIZ MCQ

C05-K05-Q-M02

The addition-theorem value $\sin 75°$ is:

A $(\sqrt{6}+\sqrt{2})/4$

B $(\sqrt{6}-\sqrt{2})/4$

C $1/2$

D 1

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | C05-K05-F09-VQUIZ

Quiz M3 | QUIZ MCQ

C05-K05-Q-M03

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

A -2

B -1

C 1

D 2

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | C05-K05-F03-VQUIZ

Quiz M4 | QUIZ MCQ

C05-K05-Q-M04

The angle between slopes 2 and -3 is:

A $\pi/6$

B $\pi/4$

C $\pi/3$

D $\pi/2$

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t - h)^2 + k$$

Addition

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta + \alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | C05-K05-F01-VQUIZ

Quiz W1 | QUIZ WRITTEN

C05-K05-Q-W01

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$ on $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | C05-K05-F04-VQUIZ

Quiz W2 | QUIZ WRITTEN

C05-K05-Q-W02

Find $\sin 2\alpha,\ \cos 2\alpha,\ \tan 2\alpha$ when $\sin\alpha = 1/3$ and α is in QII.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F07 | C05-K05-F07-VQUIZ

Quiz W3 | QUIZ WRITTEN

C05-K05-Q-W03

Rewrite $3\sin\theta - \sqrt{3}\cos\theta$ as $r\sin(\theta+\alpha)$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | C05-K05-F08-VQUIZ

Quiz W4 | QUIZ WRITTEN

C05-K05-Q-W04

Solve $\sin\theta \le \cos\theta$ on $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$